Precision Oncology Actionable Packet

Professional summary for clinicians and patients

Purpose of this packet

- Translate the uploaded review packet into a concise, readable clinical summary.
- Separate high-value biology from noise, pharmacogenomics, and technical limitations.
- Show cause and effect, treatment logic, follow-up tests, and key uncertainties.
- This is an educational support document and not a substitute for physician judgment.

Source	Uploaded Translume review packet
Report type	NGS
Disease / tumor type	Not clearly surfaced by the packet; this remains a major limitation for population fit.
Specimen context	Not consistently surfaced. Expression pages appear to reference a soft tissue / chest wall specimen.
Overall validation status	Needs Review
Important note	Automated reasoning artifacts in the packet contain some noisy or misclassified fields. This report filters those outputs into clinically meaningful categories.

1. Executive Summary

The most important biology in the uploaded packet is CHEK2 loss-of-function, along with MTAP loss and CDKN2A loss. Taken together, these findings suggest two broad lines of thinking: (1) a DNA-damage-response vulnerability centered on CHEK2, and (2) a 9p21 / MTAP–CDKN2A vulnerability that creates interest in PRMT5 / MAT2A-pathway therapies, primarily through clinical trials.

The packet also includes a clearly important technical limitation: TAF1 low coverage. That does not mean TAF1 is normal. It means the assay did not read that region well enough to make a reliable call.

A major practical limitation is that the packet does not clearly resolve tumor type, disease setting, stage, line of therapy, or prior-treatment context. Because of that, the biologic rationale may be real while the clinical fit remains unresolved.

Top takeaway	Best immediate conversation: review CHEK2-driven DDR logic and MTAP-directed trial opportunities.
Most trial-relevant finding	MTAP loss / underexpression, especially with CDKN2A loss.
Most likely to require confirmation	CHEK2 interpretation, MTAP loss confirmation, tumor type / stage / prior-therapy alignment.
Most important technical caveat	TAF1 low coverage means the assay could not confidently evaluate that region.

2. Key Findings at a Glance

These are the findings most worth discussing in a tumor-board or clinician-patient conversation.

Marker / finding	Reasoning domain	What it means	Why it matters	Actionability	Next step
CHEK2 splice-region loss-of-function	Tumor biology / therapeutic biomarker	A likely damaging CHEK2 alteration was extracted from the report and is the clearest high-priority finding in the packet.	CHEK2 is a DNA-damage-response tumor suppressor. Loss of function can weaken DNA repair and create rationale for DNA-damage-response strategies.	Potentially actionable, but context-dependent and needs clinician review.	Confirm pathogenicity, tumor context, and whether this is somatic vs germline-relevant as clinically appropriate.
MTAP loss / underexpression	Tumor biology / therapeutic biomarker	MTAP loss was extracted and is biologically important, especially because it often co-occurs with CDKN2A loss.	MTAP loss can create a synthetic-lethal vulnerability in the PRMT5/MAT2A pathway.	Actionable mainly for clinical-trial strategy.	Confirm by IHC and/or copy-number review if needed; assess MTAP-directed trial eligibility.
CDKN2A loss / underexpression	Tumor biology	CDKN2A appears reduced or lost.	CDKN2A is a cell-cycle regulator. Loss can remove brakes on tumor growth and often travels with MTAP loss on 9p21.	Biologically important; may inform prognosis and pathway logic more than it directly selects standard therapy.	Correlate with tumor type and consider p16/CDKN2A confirmation if management would change.
TAF1 low coverage	Technical limitation	This is not a mutation call. It means the assay did not read the region well enough to rule a mutation in or out.	A negative result for TAF1 cannot be treated as reassuring because the region was not adequately covered.	Not biologically actionable.	Repeat or orthogonally test only if TAF1 becomes clinically relevant.

Other findings worth noting

Finding	Interpretation
TP53 splice-region variant (LOF)	Potential secondary tumor suppressor finding; needs confirmation and tumor-context interpretation.
AKT2 overexpression	Suggests possible PI3K/AKT pathway activity, but by itself is exploratory and not treatment-directing.
RPS15 p.Y44C	Variant of unknown significance; monitor but do not anchor therapy decisions on it alone.
LYN copy-number gain	Exploratory pathway signal only; insufficient by itself to select therapy.
CDKN2B discordant signals	The packet contains both copy-number loss and RNA overexpression signals; treat as unresolved until reviewed.
TPMT variants	Best viewed as pharmacogenomic/non-tumor information rather than direct tumor targeting.

Finding	Interpretation
HLA-C genotype	Contextual immune/genotyping information, not a standalone tumor driver.

3. Cause and Effect: Why These Findings Matter

CHEK2 loss-of-function → weaker DNA-damage checkpoint control → greater dependence on backup repair pathways → rationale for DNA-damage-response strategies such as PARP- or ATR-oriented thinking.

MTAP loss → altered methylthioadenosine (MTA) / methylation balance → relative PRMT5 pathway vulnerability → rationale for MTA-cooperative PRMT5 inhibitors and MAT2A inhibitors.

CDKN2A loss → loss of cell-cycle braking (p16 / ARF axis) → easier tumor proliferation → supports a cell-cycle dysregulation interpretation and helps explain why the 9p21 region matters.

TAF1 low coverage → insufficient sequencing depth → the assay cannot confidently say whether TAF1 is altered or not.

Plain-English version

- CHEK2 suggests the tumor may be worse at repairing DNA damage.
- MTAP loss creates a possible weak spot that certain newer trial drugs are designed to exploit.
- CDKN2A loss supports the idea that the tumor has lost a growth-control brake.
- TAF1 was not read well enough, so no one should assume it is normal just because no mutation was reported.

4. Therapy Options and Clinical Rationale

The table below is organized around therapy logic, not just gene names. It is meant to support discussion, not to issue a treatment recommendation. For each option, the key question is whether the biology, tumor type, prior therapy, and current disease setting all line up.

Marker / target	Therapy class	Example agents	Molecular interaction / why it could work	Key caveats	Status
CHEK2 / DNA-repair vulnerability	PARP inhibitors	olaparib, talazoparib, niraparib, rucaparib	PARP inhibition blocks single-strand break repair and can trap PARP on DNA. In a tumor already impaired in DNA-damage response, this can increase lethal DNA damage.	Most compelling in BRCA/HRD settings. A CHEK2 finding alone is not the same as BRCA1/2 and does not automatically establish benefit. Tumor type and evidence alignment matter.	Context-dependent / discuss with oncology team
CHEK2 / DNA-repair vulnerability	ATR-pathway inhibitors	ceralasertib, camonsertib, elimusertib	ATR inhibition removes the cell's ability to pause and repair replication stress, which may increase synthetic lethality in DDR-deficient tumors.	Mainly investigational; best framed as a trial strategy rather than a standard-of-care recommendation.	Investigational
CHEK2 / checkpoint dependence	WEE1 or DNA-PK axis strategies	adavosertib, peposertib	These strategies force damaged cells through the cell cycle or impair double-strand break repair, increasing vulnerability when checkpoint control is already impaired.	Exploratory and trial-oriented. Requires disease-context fit and clinician review.	Investigational
MTAP loss	MTA-cooperative PRMT5 inhibitors	AMG 193, MRTX1719, TNG908, TNG462, IDE892	MTAP loss causes MTA accumulation, which partially stresses PRMT5. MTA-cooperative PRMT5 inhibitors are designed to exploit that altered state preferentially in MTAP-loss tumors.	This is one of the strongest biologic opportunities in the packet, but these agents are primarily in clinical development.	High-interest trial strategy
MTAP loss	MAT2A inhibitors	IDE397, AG-270 / S095033	MAT2A inhibition lowers SAM, the methyl donor needed by PRMT5, which can further weaken PRMT5-dependent biology in MTAP-loss tumors.	Also largely trial-based. Useful when evaluating MTAP-directed studies.	High-interest trial strategy
MTAP loss	PRMT5 + MAT2A combinations	IDE892 + IDE397; other sponsor-specific combinations	The combination aims to intensify synthetic lethality by attacking the same metabolic dependency from two directions.	Mechanistically strong, but still investigational and dependent on trial access and eligibility.	Investigational combination strategy
CDKN2A loss / cell-cycle dysregulation	CDK4/6-axis reasoning	palbociclib, ribociclib, abemaciclib	CDKN2A loss removes a cell-cycle brake and can signal cell-cycle dependence. That creates biologic interest in the CDK4/6 axis.	Important nuance: CDKN2A loss alone does not automatically justify CDK4/6 therapy outside disease settings where these drugs are established.	Biologic rationale, not stand alone indication
AKT2 overexpression	PI3K/AKT/mTOR-axis reasoning	capivasertib, ipatasertib, everolimus	AKT2 overexpression may reflect pro-survival signaling and could help frame future pathway-based reasoning.	Exploratory only unless broader pathway evidence and tumor-type fit support it.	Low-confidence exploratory

Practical readout

- The strongest trial-oriented opportunity in this packet is the MTAP / PRMT5 / MAT2A axis.
- The most clinically familiar concept is the CHEK2-related DDR discussion, but this still needs careful evidence alignment because CHEK2 is not automatically equivalent to BRCA1/2 biology.
- CDKN2A loss is important biology, but it is more useful for pathway framing than as a standalone standard-treatment selector.

5. Resistance and Escape Routes

Any treatment pressure can reshape the tumor. These are the most relevant escape themes to watch for.

- For DDR-oriented therapies: reversion or bypass repair, restoration of homologous recombination, alternative checkpoint signaling, and resistant subclone selection.
- For MTAP-directed therapies: metabolic rewiring of the MTA/SAM balance, PRMT5-pathway adaptation, splicing adaptation, or outgrowth of non-MTAP-loss subclones.
- For cell-cycle strategies: RB-pathway escape, cyclin E/CDK2 activation, or bypass growth signaling.
- Across all options: the biggest practical resistance/uncertainty issue in this packet is not just biology; it is incomplete population alignment (tumor type, stage, prior therapy, and treatment setting are not clearly resolved).

6. Follow-up Tests for Precision Oncology

Recommended next step	Why it matters
Pathology/oncology review of the source NGS report	Before acting on any automated interpretation, confirm the actual report wording, disease context, line of therapy, and whether the finding is tumor-relevant in this case.
MTAP IHC and/or orthogonal copy-number confirmation	Useful if MTAP-directed therapy or trial screening is being considered.
CDKN2A / p16 review (IHC if appropriate)	Helps confirm the cell-cycle interpretation and may refine the significance of the 9p21 region loss.
Clarify CHEK2 status	Review pathogenicity, zygosity when available, and whether somatic-only vs possible germline implications need attention.
Repeat or expanded profiling at progression	Repeat NGS/RNA profiling or ctDNA can help identify new resistance mechanisms if the disease evolves.
TAF1 orthogonal testing only if clinically relevant	Because TAF1 was low coverage, additional testing is warranted only if that gene becomes clinically important to management.

7. Future Phenotypic Events to Watch For

These are the re-testing triggers that matter most if the disease evolves.

Clinical event	Why it matters	Urgency
Radiographic progression	Re-biopsy or repeat profiling can reveal resistance and change treatment direction.	High
Mixed response or oligoprogression	Different tumor sites may be evolving differently; molecular re-check can be useful.	Medium
Rapid progression on targeted therapy	Suggests escape biology; reassess the tumor rather than assuming the original state is unchanged.	High
Rising tumor markers before imaging progression	May be an early sign that the phenotype is shifting.	Medium
New metastatic site	A new site can carry new biology and may justify repeat tissue or liquid profiling.	High
Before switching systemic therapy	Fresh molecular context can improve the next treatment decision.	Medium
ctDNA-negative progression	Negative ctDNA does not exclude evolving disease biology; tissue reassessment may still be needed.	Medium
Suspected histologic transformation	This is a high-priority trigger for repeat pathology and molecular review.	High

8. Limitations and Confidence Boundaries

- The packet does not clearly resolve tumor type, stage, line of therapy, or prior-treatment history. That weakens patient-population alignment for any therapy claim.
- The uploaded automated packet contains clear noise in some extracted entities and pathway labels. This report filters those artifacts rather than repeating them uncritically.
- TAF1 was low coverage. A non-call there should not be interpreted as a true negative.
- The packet indicates that assay sensitivity and clinical interpretation limits apply; not every clinically relevant variant may have been detected.
- The Medea literature reasoning component timed out in the source packet, so unsupported claims should remain conservative and clinician-reviewed.

9. Selected Reference and Trial Links

The links below are the most relevant starting points for MTAP / PRMT5 / MAT2A follow-up.

Title	URL	Why it is useful
AMG 193 Shows Preliminary Clinical Activity in MTAP-Deleted Solid Tumors	https://www.onclive.com/view/amg-193-shows-preliminary-clinical-activity-in-mtap-deleted-solid-tumors	Early clinical summary of an MTA-cooperative PRMT5 inhibitor in MTAP-deleted solid tumors.
AMG 193, a Clinical Stage MTA-Cooperative PRMT5 Inhibitor	https://pmc.ncbi.nlm.nih.gov/articles/PMC11726016/	NIH/PMC full text on discovery and early clinical evidence for AMG 193.

Title	URL	Why it is useful
TNG908 is a brain-penetrant, MTA-cooperative PRMT5 inhibitor	https://www.sciencedirect.com/science/article/pii/S1936523324003905	Primary paper describing the mechanism and MTAP-null selectivity of TNG908.
Discovery of TNG908	https://pubs.acs.org/doi/10.1021/acs.jmedchem.4c00133	Medicinal chemistry paper on TNG908 discovery.
MAT2A inhibitor AG-270/S095033 in patients with advanced malignancies	https://www.nature.com/articles/s41467-024-55316-5	Phase 1 clinical paper on AG-270 / S095033.
Combined inhibition by PRMT5 and MAT2A demonstrates synthetic lethality	https://www.nature.com/articles/s41420-025-02545-2	Preclinical/translation rationale for combining PRMT5 and MAT2A inhibition.
Study of AG-270 in Participants With Advanced Solid Tumors or Lymphoma	https://clinicaltrials.gov/study/NCT03435250	ClinicalTrials.gov record for AG-270 / S095033.
Phase 1 Study of MRTX1719 in Solid Tumors With MTAP Deletion	https://clinicaltrials.gov/study/NCT05245500	ClinicalTrials.gov record for MRTX1719 in MTAP-deleted solid tumors.
IDE397 trial	https://clinicaltrials.gov/study/NCT04794699	ClinicalTrials.gov record for IDE397.
IDE892 trial	https://clinicaltrials.gov/study/NCT07277413	ClinicalTrials.gov record for IDE892 monotherapy/combination therapy in MTAP-deleted tumors.

Bottom line

- This packet is most useful when it helps the care team move from a raw mutation list to a structured conversation about biology, therapeutic fit, confirmatory testing, resistance, and trial access.
- If only one area is pursued aggressively, the MTAP / PRMT5 / MAT2A axis is the most differentiated opportunity surfaced by the uploaded material.

10. URL-by-URL Candidate Fit Assessments

Appendix added to Precision_Oncology_Actionable_Packet_v2.pdf. Existing report pages are preserved exactly; this appendix adds the requested candidate-fit structure for every URL in the v2 reference table, plus the Panome Bio URL used as the detailed example.

Data basis and interpretation rules

- Translume review packet is treated as the actionable data source; the NGS PDF is used as ground truth to check the extracted findings.
- Patient-specific anchor: MTAP loss/underexpression, CDKN2A loss/underexpression, CDKN2B loss signal, CHEK2 loss-of-function, and TAF1 low coverage.
- Every URL section separates biologic rationale from actual treatment readiness.
- No section below is a treatment recommendation. The correct clinical use is tumor-board discussion, confirmatory testing, and clinical-trial screening when appropriate.

Scoring guide: 9–10 = very strong fit for trial screening/tumor-board review; 7–8 = strong molecular rationale but requires validation; 4–6 = plausible but limited by tumor-type or evidence gaps; 1–3 = weak or not clinically actionable.

#	Source	Marker / pathway	Evidence type
10.1	Panome Bio — Metabolomic Logic Behind MAT2A/PRMT5 Combinations	MTAP / PRMT5 / MAT2A	Mechanistic / translational metabolomics commentary
10.2	OncLive — AMG 193 Shows Preliminary Clinical Activity in MTAP-Deleted Solid Tumors	MTAP / PRMT5 / AMG 193	Clinical news summary of early AMG 193 phase 1 activity
10.3	NIH/PMC — AMG 193, a Clinical Stage MTA-Cooperative PRMT5 Inhibitor	MTAP / PRMT5 / AMG 193	Mechanistic and early clinical-development paper
10.4	ScienceDirect — TNG908 is a Brain-Penetrant, MTA-Cooperative PRMT5 Inhibitor	MTAP / PRMT5 / TNG908	Drug mechanism / MTAP-null selectivity paper
10.5	ACS — Discovery of TNG908	MTAP / PRMT5 / TNG908	Medicinal chemistry / drug discovery paper
10.6	Nature Communications — MAT2A Inhibitor AG-270/S095033 in Advanced Malignancies	MTAP / MAT2A / AG-270-S095033	Phase 1 MAT2A inhibitor clinical study
10.7	Nature / Cell Death Discovery — Combined Inhibition by PRMT5 and MAT2A Demonstrates Synthetic Lethality	MTAP / PRMT5 + MAT2A combination	Preclinical combination evidence
10.8	ClinicalTrials.gov — NCT03435250: AG-270/S095033	MTAP / MAT2A / AG-270-S095033	Clinical trial record for MAT2A inhibition
10.9	ClinicalTrials.gov — NCT05245500: MRTX1719 in Solid Tumors With MTAP Deletion	MTAP / PRMT5 / MRTX1719	Clinical trial record for an MTA-cooperative PRMT5 inhibitor
10.10	ClinicalTrials.gov — NCT04794699: IDE397	MTAP / MAT2A / IDE397	Clinical trial record for IDE397 MAT2A inhibition
10.11	ClinicalTrials.gov — NCT07277413: IDE892 Monotherapy and Combination in MTAP-Deleted Advanced Solid Tumors	MTAP / PRMT5 + MAT2A / IDE892 + IDE397	Clinical trial record for IDE892 and IDE397 combination strategy

10.1. Panome Bio — Metabolomic Logic Behind MAT2A/PRMT5 Combinations

URL	https://panomebio.com/blog/metabolomic-logic-behind-mat2a-prmt5-combinations/
Source type	Mechanistic / translational metabolomics commentary
Relevant marker/path way	MTAP / PRMT5 / MAT2A
Opening assessment	Based on the NGS findings, I would call this a strong molecular and trial-screening candidate, but not a proven treatment candidate outside a clinical trial.

Bottom-line score

Question	Strength	Why
Does the tumor biology match the Panome article's MAT2A/PRMT5 logic?	Strong: 8/10	The packet shows MTAP loss/underexpression and CDKN2A loss/underexpression, consistent with the MTAP-loss biology that makes MAT2A/PRMT5 strategies relevant.
Is this clinically actionable today as standard care?	Weak: 2/10	These are investigational agents/combos, not routine approved therapy.
Is this strong enough to justify trial screening / tumor-board discussion?	Strong: 8–9/10	Yes. The MTAP signal should trigger serious review of MTAP-directed trials, confirmatory testing, and eligibility.
Would Panome-style metabolomics be useful?	Moderate-to-strong: 7/10	Useful for pathway engagement and resistance monitoring, but not a replacement for MTAP confirmation and trial matching.

Why the fit is strong

The article's core claim is that MTAP-deleted tumors accumulate MTA, and accumulated MTA partially inhibits PRMT5. MAT2A inhibition lowers SAM, the methyl donor PRMT5 needs, creating a metabolic pincer effect against PRMT5-dependent tumor survival.

The case has the right signal: MTAP underexpression/loss, MTAP copy-number loss, and CDKN2A loss/underexpression. The NGS report also surfaced MTAP-directed trial context involving TNG908 and AMG 193.

MTAP and CDKN2A sit together in the 9p21 region, so CDKN2A loss makes the MTAP finding more believable as a regional co-deletion pattern requiring confirmation.

What would make him a stronger candidate

- MTAP loss is confirmed by MTAP IHC and/or clean copy-number evidence showing homozygous deletion or true protein loss.
- CDKN2A/p16 loss supports the 9p21 deletion pattern.
- Clinical status shows advanced, metastatic, refractory, or standard-option-limited disease.
- Trial eligibility matches cancer type, organ function, performance status, prior therapy, and protocol requirements.

What weakens the case

- The weakest part is clinical evidence maturity. The source is a mechanism/metabolomics explainer, not a clinical trial proving benefit in this patient.
- Metabolomics can guide pathway engagement and resistance monitoring, but it does not replace confirmatory pathology or trial eligibility review.

My read on this case

This was not a ‘maybe irrelevant’ finding. It was a high-priority precision-oncology lead. The right framing is: confirm MTAP protein loss, then evaluate eligibility for MTAP-directed trials involving MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, or PRMT5/MAT2A combinations.

Clinical framing to use

> This tumor has an MTAP/CDKN2A-loss pattern consistent with a 9p21 vulnerability. Confirm MTAP protein loss, then evaluate eligibility for MTAP-directed trials involving MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, or PRMT5/MAT2A combinations.

Do not say: ‘This patient should definitely receive MAT2A/PRMT5 therapy.’

Say: ‘This patient is a strong candidate for MTAP-directed trial screening, and the NGS result is strong enough that an oncologist should not ignore it.’

What should have been ordered or checked next

Follow-up	Why
MTAP IHC	Confirms whether MTAP protein is actually lost and whether the PRMT5/MAT2A trial biology is functionally present.
CDKN2A / p16 IHC or copy-number review	Supports the suspected 9p21 co-deletion biology and cell-cycle-loss interpretation.
Clean CNV review of MTAP / CDKN2A / CDKN2B	Distinguishes true regional 9p21 loss from noisy expression-only inference.
Trial search for MTAP-deleted advanced solid tumors	Especially MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, and PRMT5 + MAT2A combinations.
Metabolomic profiling if available through trial/research	Useful for SAM/MTA pathway engagement and resistance tracking; not usually a standard clinical prerequisite.

Final judgment

Source-specific conclusion

- Strong precision-oncology candidate; strong MAT2A/PRMT5 clinical-trial candidate; not a standard-of-care treatment claim.
- The report is strongest as a precision-oncology trial-screening document. It does not establish a standard-of-care treatment recommendation.

10.2. OncLive — AMG 193 Shows Preliminary Clinical Activity in MTAP-Deleted Solid Tumors

URL	https://www.onclive.com/view/amg-193-shows-preliminary-clinical-activity-in-mtap-deleted-solid-tumors
Source type	Clinical news summary of early AMG 193 phase 1 activity
Relevant marker/path way	MTAP / PRMT5 / AMG 193
Opening assessment	Based on the NGS findings, I would call this a strong MTAP-directed clinical-trial lead, but not a standard-care treatment option.

Bottom-line score

Question	Strength	Why
Does the tumor biology match AMG 193 logic?	Strong: 8/10	AMG 193 is an MTA-cooperative PRMT5 inhibitor for MTAP-deleted tumors, and the packet shows MTAP loss plus CDKN2A loss.
Is this clinically actionable today as standard care?	Weak: 2/10	The source reports early-phase clinical activity, not an approved standard treatment.
Is this strong enough for trial screening / tumor-board discussion?	Strong: 8/10	Yes. AMG 193 appears directly in the NGS report's MTAP-null trial context.
Is the tumor-type evidence perfectly aligned?	Moderate: 5/10	The reported activity spans several solid tumors, but dedifferentiated chondrosarcoma is not clearly a highlighted subgroup.

Why the fit is strong

AMG 193 targets the vulnerability implied by MTAP loss: MTA accumulation and PRMT5 pathway stress.

The NGS report already listed an AMG 193 MTAP-null trial context, making this source more directly case-relevant than a generic literature search.

The case's MTAP loss and CDKN2A loss make the mechanism biologically coherent and justify confirmatory testing.

What would make him a stronger candidate

- MTAP protein loss is confirmed centrally or by trial-acceptable IHC.
- The tumor is advanced/metastatic or lacks good standard options.
- The trial accepts dedifferentiated chondrosarcoma or broad solid tumors and the patient meets ECOG, organ function, and biopsy criteria.
- Copy-number review confirms homozygous MTAP or CDKN2A/MTAP deletion rather than expression-only inference.

What weakens the case

- Early response data do not guarantee benefit in dedifferentiated chondrosarcoma.
- Trial access, site availability, prior therapy, and eligibility would govern practical use.
- This is a clinical-development signal, not a direct treatment instruction.

My read on this case

This source supports saying: this patient should have been screened for MTAP-directed PRMT5 inhibitor trials if clinically appropriate. I would not say AMG 193 should definitely be given outside a trial.

Clinical framing to use

> This tumor has an MTAP/CDKN2A-loss pattern consistent with a 9p21 vulnerability. Confirm MTAP protein loss, then evaluate eligibility for MTAP-directed trials involving MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, or PRMT5/MAT2A combinations.

Do not say: ‘This patient should definitely receive MAT2A/PRMT5 therapy.’

Say: ‘This patient is a strong candidate for MTAP-directed trial screening, and the NGS result is strong enough that an oncologist should not ignore it.’

What should have been ordered or checked next

Follow-up	Why
MTAP IHC	Confirms whether MTAP protein is actually lost and whether the PRMT5/MAT2A trial biology is functionally present.
CDKN2A / p16 IHC or copy-number review	Supports the suspected 9p21 co-deletion biology and cell-cycle-loss interpretation.
Clean CNV review of MTAP / CDKN2A / CDKN2B	Distinguishes true regional 9p21 loss from noisy expression-only inference.
Trial search for MTAP-deleted advanced solid tumors	Especially MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, and PRMT5 + MAT2A combinations.
Metabolomic profiling if available through trial/research	Useful for SAM/MTA pathway engagement and resistance tracking; not usually a standard clinical prerequisite.
Confirm advanced/metastatic/refractory setting	Most early-phase MTAP-directed trials require advanced, unresectable, metastatic, recurrent, or standard-option-exhausted disease.
Check trial inclusion/exclusion criteria	Eligibility depends on tumor type, prior therapy, ECOG performance status, organ function, measurable disease, and biopsy feasibility.
Confirm MTAP loss with acceptable assay	Many trials require homozygous MTAP deletion and/or MTAP protein loss by IHC, not just a noisy expression signal.

Final judgment

Source-specific conclusion

- Strong clinical-trial lead if MTAP loss is confirmed; not a stand-alone treatment recommendation.
- The report is strongest as a precision-oncology trial-screening document. It does not establish a standard-of-care treatment recommendation.

10.3. NIH/PMC — AMG 193, a Clinical Stage MTA-Cooperative PRMT5 Inhibitor

URL	https://pmc.ncbi.nlm.nih.gov/articles/PMC11726016/
Source type	Mechanistic and early clinical-development paper
Relevant marker/path way	MTAP / PRMT5 / AMG 193
Opening assessment	Based on the NGS findings, I would call this strong mechanistic support for AMG 193 trial-screening, but not proof of patient-specific response.

Bottom-line score

Question	Strength	Why
Does the tumor biology match the paper's mechanism?	Strong: 8/10	The paper's mechanism requires MTAP loss / MTA accumulation, which is supported by the packet.
Is it clinically actionable today as standard care?	Weak: 2/10	Clinical-stage drug-development evidence is not routine clinical care.
Does it support tumor-board discussion?	Strong: 8/10	Yes. It explains why an MTA-cooperative PRMT5 inhibitor is rational in MTAP-loss tumors.
Does it prove this patient would benefit?	Weak-to-moderate: 4/10	It supports mechanism and early signal, but not individual response prediction.

Why the fit is strong

This source explains why AMG 193 was designed to preferentially exploit the MTAP-deleted / MTA-accumulated state rather than inhibit PRMT5 broadly in all cells.

The patient's MTAP loss is the exact enabling biomarker; CDKN2A/CDKN2B loss makes broader 9p21-region biology more plausible.

Because the NGS report also listed an AMG 193 trial context, the source should be tied directly to trial screening rather than left as background science.

What would make him a stronger candidate

- Clean CNV shows homozygous MTAP deletion or paired MTAP/CDKN2A loss.
- MTAP IHC shows loss of protein expression.
- Trial criteria accept the disease setting and do not exclude prior treatment history or organ-function profile.
- The oncology team can access a trial site or compassionate-use path only through appropriate regulatory/clinical channels.

What weakens the case

- The paper does not resolve clinical eligibility, tumor subtype, prior therapy, organ function, or biopsy feasibility.
- Mechanistic fit is not the same as proven response in dedifferentiated chondrosarcoma.

My read on this case

This should be used as clinician-facing mechanism support for MTAP-directed trial review. It should not be used to claim AMG 193 is proven therapy for this individual patient.

Clinical framing to use

> This tumor has an MTAP/CDKN2A-loss pattern consistent with a 9p21 vulnerability. Confirm MTAP protein loss, then evaluate eligibility for MTAP-directed trials involving MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, or

PRMT5/MAT2A combinations.

Do not say: ‘This patient should definitely receive MAT2A/PRMT5 therapy.’

Say: ‘This patient is a strong candidate for MTAP-directed trial screening, and the NGS result is strong enough that an oncologist should not ignore it.’

What should have been ordered or checked next

Follow-up	Why
MTAP IHC	Confirms whether MTAP protein is actually lost and whether the PRMT5/MAT2A trial biology is functionally present.
CDKN2A / p16 IHC or copy-number review	Supports the suspected 9p21 co-deletion biology and cell-cycle-loss interpretation.
Clean CNV review of MTAP / CDKN2A / CDKN2B	Distinguishes true regional 9p21 loss from noisy expression-only inference.
Trial search for MTAP-deleted advanced solid tumors	Especially MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, and PRMT5 + MAT2A combinations.
Metabolomic profiling if available through trial/research	Useful for SAM/MTA pathway engagement and resistance tracking; not usually a standard clinical prerequisite.
Confirm advanced/metastatic/refractory setting	Most early-phase MTAP-directed trials require advanced, unresectable, metastatic, recurrent, or standard-option-exhausted disease.
Check trial inclusion/exclusion criteria	Eligibility depends on tumor type, prior therapy, ECOG performance status, organ function, measurable disease, and biopsy feasibility.
Confirm MTAP loss with acceptable assay	Many trials require homozygous MTAP deletion and/or MTAP protein loss by IHC, not just a noisy expression signal.

Final judgment

Source-specific conclusion

- Strong mechanistic support for trial screening; not sufficient as proof of benefit.
- The report is strongest as a precision-oncology trial-screening document. It does not establish a standard-of-care treatment recommendation.

10.4. ScienceDirect — TNG908 is a Brain-Penetrant, MTA-Cooperative PRMT5 Inhibitor

URL	https://www.sciencedirect.com/science/article/pii/S1936523324003905
Source type	Drug mechanism / MTAP-null selectivity paper
Relevant marker/pathway	MTAP / PRMT5 / TNG908
Opening assessment	Based on the NGS findings, I would call this a strong TNG908 trial-screening lead if MTAP loss is confirmed, but not a standard-care treatment option.

Bottom-line score

Question	Strength	Why
Does the tumor biology match TNG908's intended biomarker?	Strong: 8/10	TNG908 is designed around MTAP-null, MTA-cooperative PRMT5 inhibition, and the packet shows MTAP loss.
Is brain penetration relevant here?	Unclear: 3/10	Brain penetration matters for CNS disease; this ground-truth report points to soft tissue/chest wall disease, not CNS involvement.
Is this enough for trial screening?	Strong: 8/10	Yes. The NGS report itself listed a TNG908 MTAP-deleted solid tumor trial context.
Is it standard care?	Weak: 1–2/10	TNG908 is investigational.

Why the fit is strong

TNG908's mechanism aligns with MTAP loss. The key case-specific point is MTAP deletion, not brain delivery.

The paper describes MTAP-deleted selectivity and Phase I/II clinical testing in MTAP-deleted solid tumors, supporting the relevance of this trial lead.

The fit is strengthened because the NGS report itself identified a TNG908 trial context.

What would make him a stronger candidate

- MTAP protein loss is confirmed by IHC.
- Trial accepts dedifferentiated chondrosarcoma or broad MTAP-deleted solid tumors.
- Clinical status allows trial enrollment and required monitoring/biopsy procedures.
- No CNS-specific requirement is imposed unless disease distribution warrants it.

What weakens the case

- The brain-penetrant feature may not matter for chest wall/sternal disease unless CNS involvement develops.
- Preclinical and early clinical-development rationale does not establish benefit for this tumor type.
- Trial availability and cohort structure must be checked in real time.

My read on this case

This is a high-priority MTAP-directed trial-screening source, especially because it was already surfaced by the NGS report. It is not a reason to use TNG908 outside a clinical trial.

Clinical framing to use

> This tumor has an MTAP/CDKN2A-loss pattern consistent with a 9p21 vulnerability. Confirm MTAP protein loss, then evaluate eligibility for MTAP-directed trials involving MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, or PRMT5/MAT2A combinations.

Do not say: ‘This patient should definitely receive MAT2A/PRMT5 therapy.’

Say: ‘This patient is a strong candidate for MTAP-directed trial screening, and the NGS result is strong enough that an oncologist should not ignore it.’

What should have been ordered or checked next

Follow-up	Why
MTAP IHC	Confirms whether MTAP protein is actually lost and whether the PRMT5/MAT2A trial biology is functionally present.
CDKN2A / p16 IHC or copy-number review	Supports the suspected 9p21 co-deletion biology and cell-cycle-loss interpretation.
Clean CNV review of MTAP / CDKN2A / CDKN2B	Distinguishes true regional 9p21 loss from noisy expression-only inference.
Trial search for MTAP-deleted advanced solid tumors	Especially MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, and PRMT5 + MAT2A combinations.
Metabolomic profiling if available through trial/research	Useful for SAM/MTA pathway engagement and resistance tracking; not usually a standard clinical prerequisite.
Confirm advanced/metastatic/refractory setting	Most early-phase MTAP-directed trials require advanced, unresectable, metastatic, recurrent, or standard-option-exhausted disease.
Check trial inclusion/exclusion criteria	Eligibility depends on tumor type, prior therapy, ECOG performance status, organ function, measurable disease, and biopsy feasibility.
Confirm MTAP loss with acceptable assay	Many trials require homozygous MTAP deletion and/or MTAP protein loss by IHC, not just a noisy expression signal.

Final judgment

Source-specific conclusion

- Strong MTAP-directed trial-screening candidate if MTAP loss is confirmed; brain-penetrant value depends on disease distribution.
- The report is strongest as a precision-oncology trial-screening document. It does not establish a standard-of-care treatment recommendation.

10.5. ACS — Discovery of TNG908

URL	https://pubs.acs.org/doi/10.1021/acs.jmedchem.4c00133
Source type	Medicinal chemistry / drug discovery paper
Relevant marker/path way	MTAP / PRMT5 / TNG908
Opening assessment	Based on the NGS findings, I would call this a strong mechanistic rationale source for trial screening, but not clinical proof.

Bottom-line score

Question	Strength	Why
Does the tumor biology match?	Strong: 8/10	The discovery program is built around MTAP-null selectivity, matching the packet's MTAP loss signal.
Does this provide clinical efficacy?	Weak: 2/10	Medicinal chemistry and preclinical discovery support rationale, not clinical response.
Does it support trial discussion?	Moderate-to-strong: 7/10	Yes. It explains why TNG908 exists and why MTAP loss matters.
Is it enough alone for treatment selection?	Weak: 2/10	No. It must be paired with trial criteria and clinical evidence.

Why the fit is strong

This source helps explain TNG908's design logic: selective targeting of the PRMT5/MTA state associated with MTAP loss.

The fit is strong at the biomarker-mechanism level because this case has MTAP loss and supporting CDKN2A/CDKN2B loss biology.

It provides the chemical-design rationale behind a trial option, not the final clinical decision.

What would make him a stronger candidate

- MTAP loss is confirmed by trial-acceptable assay.
- TNG908 trial cohort is open for the patient's tumor type or broad solid tumors.
- Clinical trial status, geography, performance status, and organ function are acceptable.
- The treating team reviews whether current trial endpoints and toxicity profile fit the patient's goals.

What weakens the case

- Drug-discovery papers do not resolve whether the patient can receive the drug or whether the disease setting fits.
- Clinical efficacy must come from trials, not chemistry design alone.

My read on this case

This belongs in the report as rationale explaining why TNG908 is biologically relevant. It should be weighted below human clinical trial evidence.

Clinical framing to use

> This tumor has an MTAP/CDKN2A-loss pattern consistent with a 9p21 vulnerability. Confirm MTAP protein loss, then evaluate eligibility for MTAP-directed trials involving MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, or PRMT5/MAT2A combinations.

Do not say: 'This patient should definitely receive MAT2A/PRMT5 therapy.'

Say: ‘This patient is a strong candidate for MTAP-directed trial screening, and the NGS result is strong enough that an oncologist should not ignore it.’

What should have been ordered or checked next

Follow-up	Why
MTAP IHC	Confirms whether MTAP protein is actually lost and whether the PRMT5/MAT2A trial biology is functionally present.
CDKN2A / p16 IHC or copy-number review	Supports the suspected 9p21 co-deletion biology and cell-cycle-loss interpretation.
Clean CNV review of MTAP / CDKN2A / CDKN2B	Distinguishes true regional 9p21 loss from noisy expression-only inference.
Trial search for MTAP-deleted advanced solid tumors	Especially MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, and PRMT5 + MAT2A combinations.
Metabolomic profiling if available through trial/research	Useful for SAM/MTA pathway engagement and resistance tracking; not usually a standard clinical prerequisite.
Confirm advanced/metastatic/refractory setting	Most early-phase MTAP-directed trials require advanced, unresectable, metastatic, recurrent, or standard-option-exhausted disease.
Check trial inclusion/exclusion criteria	Eligibility depends on tumor type, prior therapy, ECOG performance status, organ function, measurable disease, and biopsy feasibility.
Confirm MTAP loss with acceptable assay	Many trials require homozygous MTAP deletion and/or MTAP protein loss by IHC, not just a noisy expression signal.

Final judgment

Source-specific conclusion

- Strong rationale source; not clinical proof.
- The report is strongest as a precision-oncology trial-screening document. It does not establish a standard-of-care treatment recommendation.

10.6. Nature Communications — MAT2A Inhibitor AG-270/S095033 in Advanced Malignancies

URL	https://www.nature.com/articles/s41467-024-55316-5
Source type	Phase 1 MAT2A inhibitor clinical study
Relevant marker/pathway	MTAP / MAT2A / AG-270-S095033
Opening assessment	Based on the NGS findings, I would call this a strong MAT2A trial-screening rationale, but not a proven treatment option.

Bottom-line score

Question	Strength	Why
Does the case match the biomarker logic?	Strong: 8/10	The study is relevant to MTAP/CDKN2A-deleted or MTAP-protein-loss tumors, and the packet shows MTAP/CDKN2A loss signals.
Does it show standard-care efficacy?	Weak: 2/10	Phase 1 evidence supports investigation and mechanism, not a standard treatment.
Is trial-screening justified?	Strong: 8/10	Yes. This is directly relevant to MAT2A inhibition in MTAP-loss biology.
How mature is the clinical signal?	Modest: 4–5/10	The trial supports proof of mechanism, but response rates were limited; it is not definitive efficacy evidence.

Why the fit is strong

This source is highly relevant because it ties MAT2A inhibition to the same MTAP/CDKN2A-deleted biology seen in the packet.

The study required homozygous CDKN2A/MTAP deletion and/or MTAP protein loss by IHC, which directly informs what confirmatory testing should have been ordered.

The mechanism is coherent: MAT2A inhibition lowers SAM and can deepen PRMT5 vulnerability in MTAP-loss tumors.

What would make him a stronger candidate

- MTAP protein loss is demonstrated by IHC.
- CNV review supports homozygous deletion of MTAP and/or CDKN2A/MTAP.
- Clinical status matches advanced malignancy trial criteria and prior therapy requirements.
- The patient can tolerate oral investigational therapy and required monitoring.

What weakens the case

- Clinical activity was preliminary; the study showed proof-of-mechanism with limited objective responses.
- MAT2A monotherapy may be less compelling than combination strategies in some settings.
- This does not prove benefit in dedifferentiated chondrosarcoma.

My read on this case

This source should have pushed the team toward MTAP IHC/CNV confirmation and MAT2A/PRMT5 trial screening. It should not be framed as standard therapy.

Clinical framing to use

> This tumor has an MTAP/CDKN2A-loss pattern consistent with a 9p21 vulnerability. Confirm MTAP protein loss, then evaluate eligibility for MTAP-directed trials involving MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, or PRMT5/MAT2A combinations.

Do not say: ‘This patient should definitely receive MAT2A/PRMT5 therapy.’

Say: ‘This patient is a strong candidate for MTAP-directed trial screening, and the NGS result is strong enough that an oncologist should not ignore it.’

What should have been ordered or checked next

Follow-up	Why
MTAP IHC	Confirms whether MTAP protein is actually lost and whether the PRMT5/MAT2A trial biology is functionally present.
CDKN2A / p16 IHC or copy-number review	Supports the suspected 9p21 co-deletion biology and cell-cycle-loss interpretation.
Clean CNV review of MTAP / CDKN2A / CDKN2B	Distinguishes true regional 9p21 loss from noisy expression-only inference.
Trial search for MTAP-deleted advanced solid tumors	Especially MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, and PRMT5 + MAT2A combinations.
Metabolomic profiling if available through trial/research	Useful for SAM/MTA pathway engagement and resistance tracking; not usually a standard clinical prerequisite.
Confirm advanced/metastatic/refractory setting	Most early-phase MTAP-directed trials require advanced, unresectable, metastatic, recurrent, or standard-option-exhausted disease.
Check trial inclusion/exclusion criteria	Eligibility depends on tumor type, prior therapy, ECOG performance status, organ function, measurable disease, and biopsy feasibility.
Confirm MTAP loss with acceptable assay	Many trials require homozygous MTAP deletion and/or MTAP protein loss by IHC, not just a noisy expression signal.

Final judgment

Source-specific conclusion

- Strong MAT2A trial-screening rationale; not a proven treatment claim.
- The report is strongest as a precision-oncology trial-screening document. It does not establish a standard-of-care treatment recommendation.

10.7. Nature / Cell Death Discovery — Combined Inhibition by PRMT5 and MAT2A Demonstrates Synthetic Lethality

URL	https://www.nature.com/articles/s41420-025-02545-2
Source type	Preclinical combination evidence
Relevant marker/path way	MTAP / PRMT5 + MAT2A combination
Opening assessment	Based on the NGS findings, I would call this strong pathway rationale for combination trials, but not direct clinical evidence for this patient.

Bottom-line score

Question	Strength	Why
Does the biology match?	Moderate-to-strong: 7/10	The MTAP-loss logic matches, but the study is glioma-model focused rather than chondrosarcoma-specific.
Does it support standard care?	Weak: 1/10	This is preclinical/translational combination evidence.
Does it support a trial-combination discussion?	Strong: 8/10	Yes. It supports why PRMT5 + MAT2A combinations are rational in MTAP-deficient settings.
Does it prove this patient's tumor would respond?	Weak: 2/10	No. It provides mechanism, not patient-specific efficacy.

Why the fit is strong

This source supports the concept that dual pressure on PRMT5 and MAT2A can strengthen synthetic lethality in MTAP-deficient models.

The fit is strong at the pathway level because the packet shows MTAP loss and CDKN2A loss.

The source is especially useful for explaining why combinations may matter more than single-agent strategies.

What would make him a stronger candidate

- MTAP protein loss and/or homozygous deletion is confirmed.
- A clinical trial exists for PRMT5 + MAT2A in a broad MTAP-deleted solid-tumor population or the patient's disease type.
- Trial protocol allows this tumor type and the patient meets biopsy, ECOG, and organ-function criteria.
- Monitoring can include pharmacodynamic or metabolomic endpoints if offered by the trial.

What weakens the case

- The disease model is glioma, not chondrosarcoma.
- Preclinical synthetic lethality does not equal clinical efficacy.
- Combination toxicity and scheduling require human trial data.

My read on this case

This is useful as mechanistic combination support. It strengthens the rationale for PRMT5/MAT2A combination trial screening, not for off-trial treatment.

Clinical framing to use

> This tumor has an MTAP/CDKN2A-loss pattern consistent with a 9p21 vulnerability. Confirm MTAP protein loss, then evaluate eligibility for MTAP-directed trials involving MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, or PRMT5/MAT2A combinations.

Do not say: ‘This patient should definitely receive MAT2A/PRMT5 therapy.’

Say: ‘This patient is a strong candidate for MTAP-directed trial screening, and the NGS result is strong enough that an oncologist should not ignore it.’

What should have been ordered or checked next

Follow-up	Why
MTAP IHC	Confirms whether MTAP protein is actually lost and whether the PRMT5/MAT2A trial biology is functionally present.
CDKN2A / p16 IHC or copy-number review	Supports the suspected 9p21 co-deletion biology and cell-cycle-loss interpretation.
Clean CNV review of MTAP / CDKN2A / CDKN2B	Distinguishes true regional 9p21 loss from noisy expression-only inference.
Trial search for MTAP-deleted advanced solid tumors	Especially MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, and PRMT5 + MAT2A combinations.
Metabolomic profiling if available through trial/research	Useful for SAM/MTA pathway engagement and resistance tracking; not usually a standard clinical prerequisite.
Confirm advanced/metastatic/refractory setting	Most early-phase MTAP-directed trials require advanced, unresectable, metastatic, recurrent, or standard-option-exhausted disease.
Check trial inclusion/exclusion criteria	Eligibility depends on tumor type, prior therapy, ECOG performance status, organ function, measurable disease, and biopsy feasibility.
Confirm MTAP loss with acceptable assay	Many trials require homozygous MTAP deletion and/or MTAP protein loss by IHC, not just a noisy expression signal.

Final judgment

Source-specific conclusion

- Useful as mechanistic combination support; not a direct clinical recommendation.
- The report is strongest as a precision-oncology trial-screening document. It does not establish a standard-of-care treatment recommendation.

10.8. ClinicalTrials.gov — NCT03435250: AG-270/S095033

URL	https://clinicaltrials.gov/study/NCT03435250
Source type	Clinical trial record for MAT2A inhibition
Relevant marker/pathway	MTAP / MAT2A / AG-270-S095033
Opening assessment	Based on the NGS findings, I would call this a good MAT2A trial-search anchor, but likely more useful as precedent than a direct current option unless status and sites are confirmed.

Bottom-line score

Question	Strength	Why
Does the case match trial biomarker logic?	Strong: 8/10	The trial concerns advanced tumors with MTAP loss / homozygous deletion logic, and the packet shows MTAP loss.
Is it standard care?	Weak: 1/10	Clinical-trial record only; investigational therapy.
Is this specific trial a current option?	Uncertain / likely historical	The study is useful as evidence precedent, but current availability must be checked in real time.
Does it justify looking for MAT2A trials?	Strong: 8/10	Yes. It is aligned with the packet's MTAP/CDKN2A pattern.

Why the fit is strong

The trial operationalizes the same core question: does MTAP loss create vulnerability to MAT2A inhibition?

The fit is strengthened by MTAP loss and CDKN2A loss in the packet, plus the NGS report's broader MTAP trial context.

This trial also clarifies what confirmation matters: homozygous MTAP loss and/or protein loss, not only expression-level inference.

What would make him a stronger candidate

- Trial status shows recruiting or an active related MAT2A trial is available.
- MTAP protein loss or homozygous deletion is confirmed.
- The patient's clinical status matches advanced solid tumor or lymphoma criteria.
- Performance status, organ function, measurable disease, and prior therapy fit protocol criteria.

What weakens the case

- Trial record alone is not enough; status, site availability, and eligibility must be checked.
- If the trial is closed or terminated, use it as evidence precedent rather than a current option.

My read on this case

This should trigger a focused search for current MAT2A inhibitor trials, not necessarily enrollment in this exact historical trial.

Clinical framing to use

> This tumor has an MTAP/CDKN2A-loss pattern consistent with a 9p21 vulnerability. Confirm MTAP protein loss, then evaluate eligibility for MTAP-directed trials involving MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, or PRMT5/MAT2A combinations.

Do not say: 'This patient should definitely receive MAT2A/PRMT5 therapy.'

Say: 'This patient is a strong candidate for MTAP-directed trial screening, and the NGS result is strong enough that an oncologist should not ignore it.'

What should have been ordered or checked next

Follow-up	Why
MTAP IHC	Confirms whether MTAP protein is actually lost and whether the PRMT5/MAT2A trial biology is functionally present.
CDKN2A / p16 IHC or copy-number review	Supports the suspected 9p21 co-deletion biology and cell-cycle-loss interpretation.
Clean CNV review of MTAP / CDKN2A / CDKN2B	Distinguishes true regional 9p21 loss from noisy expression-only inference.
Trial search for MTAP-deleted advanced solid tumors	Especially MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, and PRMT5 + MAT2A combinations.
Metabolomic profiling if available through trial/research	Useful for SAM/MTA pathway engagement and resistance tracking; not usually a standard clinical prerequisite.
Confirm advanced/metastatic/refractory setting	Most early-phase MTAP-directed trials require advanced, unresectable, metastatic, recurrent, or standard-option-exhausted disease.
Check trial inclusion/exclusion criteria	Eligibility depends on tumor type, prior therapy, ECOG performance status, organ function, measurable disease, and biopsy feasibility.
Confirm MTAP loss with acceptable assay	Many trials require homozygous MTAP deletion and/or MTAP protein loss by IHC, not just a noisy expression signal.

Final judgment

Source-specific conclusion

- Good precedent and trial-search anchor; may not be active depending on current status.
- The report is strongest as a precision-oncology trial-screening document. It does not establish a standard-of-care treatment recommendation.

10.9. ClinicalTrials.gov — NCT05245500: MRTX1719 in Solid Tumors With MTAP Deletion

URL	https://clinicaltrials.gov/study/NCT05245500
Source type	Clinical trial record for an MTA-cooperative PRMT5 inhibitor
Relevant marker/pathway	MTAP / PRMT5 / MRTX1719
Opening assessment	Based on the NGS findings, I would call this one of the strongest URL-to-case trial-screening matches, assuming MTAP loss is confirmed and eligibility fits.

Bottom-line score

Question	Strength	Why
Does the case match trial biomarker logic?	Strong: 8–9/10	The trial is explicitly for MTAP-deleted solid tumors, and the packet shows MTAP loss.
Is it standard care?	Weak: 1/10	This is an investigational trial.
Is tumor-board/trial screening justified?	Strong: 8–9/10	Yes, if MTAP loss is confirmed and the clinical setting meets eligibility.
What is the biggest fit gap?	Eligibility context	The packet does not fully resolve stage, prior therapy, performance status, organ function, or whether MTAP loss is homozygous/protein-confirmed.

Why the fit is strong

MRTX1719 is a clean URL-to-case match because the trial title itself centers on MTAP deletion in solid tumors.

The NGS report shows MTAP copy-number loss, and the Translume packet highlights MTAP loss/underexpression, which should trigger trial screening.

The PRMT5 inhibitor mechanism is directly tied to MTAP loss/MTA accumulation.

What would make him a stronger candidate

- Homozygous MTAP deletion is confirmed in tumor tissue.
- Disease is advanced, unresectable, metastatic, or otherwise trial-eligible.
- A lesion is available for required biopsy if the protocol requires pharmacodynamic evaluation.
- ECOG performance status and organ function satisfy criteria.

What weakens the case

- Eligibility may require homozygous deletion and biopsy feasibility, not just underexpression.
- The trial may restrict cohorts, sites, prior therapies, or concurrent treatments.
- Trial access depends on current recruitment status.

My read on this case

This is a strong trial-screening candidate. The correct action would be confirmatory MTAP testing plus protocol-level eligibility review.

Clinical framing to use

> This tumor has an MTAP/CDKN2A-loss pattern consistent with a 9p21 vulnerability. Confirm MTAP protein loss, then evaluate eligibility for MTAP-directed trials involving MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, or PRMT5/MAT2A combinations.

Do not say: ‘This patient should definitely receive MAT2A/PRMT5 therapy.’

Say: ‘This patient is a strong candidate for MTAP-directed trial screening, and the NGS result is strong enough that an oncologist should not ignore it.’

What should have been ordered or checked next

Follow-up	Why
MTAP IHC	Confirms whether MTAP protein is actually lost and whether the PRMT5/MAT2A trial biology is functionally present.
CDKN2A / p16 IHC or copy-number review	Supports the suspected 9p21 co-deletion biology and cell-cycle-loss interpretation.
Clean CNV review of MTAP / CDKN2A / CDKN2B	Distinguishes true regional 9p21 loss from noisy expression-only inference.
Trial search for MTAP-deleted advanced solid tumors	Especially MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, and PRMT5 + MAT2A combinations.
Metabolomic profiling if available through trial/research	Useful for SAM/MTA pathway engagement and resistance tracking; not usually a standard clinical prerequisite.
Confirm advanced/metastatic/refractory setting	Most early-phase MTAP-directed trials require advanced, unresectable, metastatic, recurrent, or standard-option-exhausted disease.
Check trial inclusion/exclusion criteria	Eligibility depends on tumor type, prior therapy, ECOG performance status, organ function, measurable disease, and biopsy feasibility.
Confirm MTAP loss with acceptable assay	Many trials require homozygous MTAP deletion and/or MTAP protein loss by IHC, not just a noisy expression signal.

Final judgment

Source-specific conclusion

- Strong trial-screening candidate if MTAP loss is confirmed and protocol criteria fit.
- The report is strongest as a precision-oncology trial-screening document. It does not establish a standard-of-care treatment recommendation.

10.10. ClinicalTrials.gov — NCT04794699: IDE397

URL	https://clinicaltrials.gov/study/NCT04794699
Source type	Clinical trial record for IDE397 MAT2A inhibition
Relevant marker/pathway	MTAP / MAT2A / IDE397
Opening assessment	Based on the NGS findings, I would call this a strong MAT2A trial-screening lead if MTAP loss is confirmed and an eligible cohort is open.

Bottom-line score

Question	Strength	Why
Does the case match the biomarker logic?	Strong: 8/10	IDE397 targets MAT2A, directly relevant to MTAP-loss biology.
Is it standard care?	Weak: 1/10	IDE397 remains investigational.
Is screening justified?	Strong: 8/10	Yes, particularly if MTAP loss is confirmed by IHC/CNV and disease context fits an active cohort.
Disease-specific fit?	Unclear: 4–5/10	Some expansion cohorts emphasize specific tumor types; dedifferentiated chondrosarcoma fit must be checked.

Why the fit is strong

IDE397 is relevant because MAT2A inhibition lowers SAM, tightening the PRMT5 vulnerability created by MTAP loss.

The case has the right molecular cue: MTAP loss with CDKN2A/CDKN2B co-loss context.

This trial is important because MAT2A inhibitors are one of the main drug classes emerging from MTAP-loss biology.

What would make him a stronger candidate

- Confirmed homozygous MTAP loss or MTAP deletion by acceptable assay.
- Trial cohort accepts the tumor type or has a broad advanced solid tumor arm.
- Measurable disease, ECOG status, prior therapy, and biopsy requirements are satisfied.
- Combination or monotherapy arm is appropriate based on current protocol.

What weakens the case

- The biggest weakness is practical availability and cohort fit, not the biology.
- If the trial prioritizes NSCLC or urothelial cancer, chondrosarcoma inclusion may be uncertain.
- As with all investigational therapy, benefit is not guaranteed.

My read on this case

This is a high-value trial-search lead for MTAP loss. It should prompt clinical trial matching and confirmatory MTAP testing, not off-protocol treatment.

Clinical framing to use

> This tumor has an MTAP/CDKN2A-loss pattern consistent with a 9p21 vulnerability. Confirm MTAP protein loss, then evaluate eligibility for MTAP-directed trials involving MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, or PRMT5/MAT2A combinations.

Do not say: ‘This patient should definitely receive MAT2A/PRMT5 therapy.’

Say: ‘This patient is a strong candidate for MTAP-directed trial screening, and the NGS result is strong enough that an oncologist should not ignore it.’

What should have been ordered or checked next

Follow-up	Why
MTAP IHC	Confirms whether MTAP protein is actually lost and whether the PRMT5/MAT2A trial biology is functionally present.
CDKN2A / p16 IHC or copy-number review	Supports the suspected 9p21 co-deletion biology and cell-cycle-loss interpretation.
Clean CNV review of MTAP / CDKN2A / CDKN2B	Distinguishes true regional 9p21 loss from noisy expression-only inference.
Trial search for MTAP-deleted advanced solid tumors	Especially MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, and PRMT5 + MAT2A combinations.
Metabolomic profiling if available through trial/research	Useful for SAM/MTA pathway engagement and resistance tracking; not usually a standard clinical prerequisite.
Confirm advanced/metastatic/refractory setting	Most early-phase MTAP-directed trials require advanced, unresectable, metastatic, recurrent, or standard-option-exhausted disease.
Check trial inclusion/exclusion criteria	Eligibility depends on tumor type, prior therapy, ECOG performance status, organ function, measurable disease, and biopsy feasibility.
Confirm MTAP loss with acceptable assay	Many trials require homozygous MTAP deletion and/or MTAP protein loss by IHC, not just a noisy expression signal.

Final judgment

Source-specific conclusion

- Strong MAT2A trial-screening lead if cohort eligibility aligns.
- The report is strongest as a precision-oncology trial-screening document. It does not establish a standard-of-care treatment recommendation.

10.11. ClinicalTrials.gov — NCT07277413: IDE892 Monotherapy and Combination in MTAP-Deleted Advanced Solid Tumors

URL	https://clinicaltrials.gov/study/NCT07277413
Source type	Clinical trial record for IDE892 and IDE397 combination strategy
Relevant marker/path way	MTAP / PRMT5 + MAT2A / IDE892 + IDE397
Opening assessment	Based on the NGS findings, I would call this a strong precision-oncology trial-screening lead because it directly matches MTAP-deleted advanced solid tumor biology and PRMT5/MAT2A combination logic.

Bottom-line score

Question	Strength	Why
Does the case match the biomarker logic?	Strong: 8–9/10	The trial is for MTAP-deleted advanced solid tumors, and the packet shows MTAP loss.
Does it match the combination logic?	Strong: 8/10	IDE892 with IDE397 maps directly to PRMT5 + MAT2A combination rationale.
Is it standard care?	Weak: 1/10	Investigational therapy only.
Is it worth tumor-board discussion?	Strong: 8–9/10	Yes, if the patient has confirmed MTAP loss and matches disease/cohort eligibility.

Why the fit is strong

This is one of the most directly relevant URLs because it ties MTAP-deleted advanced solid tumors to PRMT5 monotherapy and PRMT5 + MAT2A combination logic.

The patient's MTAP loss, CDKN2A/CDKN2B loss, and PRMT5/MAT2A pathway fit make it a high-priority trial-screening reference.

The combination approach addresses the concern that single-agent MAT2A or PRMT5 activity may be modest in some settings.

What would make him a stronger candidate

- MTAP loss is confirmed by protein loss and/or homozygous deletion.
- Protocol accepts the patient's tumor type and treatment history.
- Disease is advanced/metastatic and standard options are exhausted, inappropriate, or ineffective.
- Patient meets age, performance status, organ function, and biopsy criteria.

What weakens the case

- The biggest caveats are current site availability, protocol inclusion/exclusion criteria, tumor-type restrictions, prior-treatment requirements, and confirmation of MTAP loss.
- Combination therapies can create added toxicity and require protocol-level monitoring.
- This remains investigational.

My read on this case

This is a strong trial-screening candidate. It should be communicated clearly as an investigational opportunity grounded in the patient's MTAP/CDKN2A loss pattern.

Clinical framing to use

> This tumor has an MTAP/CDKN2A-loss pattern consistent with a 9p21 vulnerability. Confirm MTAP protein loss, then evaluate eligibility for MTAP-directed trials involving MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, or PRMT5/MAT2A combinations.

Do not say: ‘This patient should definitely receive MAT2A/PRMT5 therapy.’

Say: ‘This patient is a strong candidate for MTAP-directed trial screening, and the NGS result is strong enough that an oncologist should not ignore it.’

What should have been ordered or checked next

Follow-up	Why
MTAP IHC	Confirms whether MTAP protein is actually lost and whether the PRMT5/MAT2A trial biology is functionally present.
CDKN2A / p16 IHC or copy-number review	Supports the suspected 9p21 co-deletion biology and cell-cycle-loss interpretation.
Clean CNV review of MTAP / CDKN2A / CDKN2B	Distinguishes true regional 9p21 loss from noisy expression-only inference.
Trial search for MTAP-deleted advanced solid tumors	Especially MTA-cooperative PRMT5 inhibitors, MAT2A inhibitors, and PRMT5 + MAT2A combinations.
Metabolomic profiling if available through trial/research	Useful for SAM/MTA pathway engagement and resistance tracking; not usually a standard clinical prerequisite.
Confirm advanced/metastatic/refractory setting	Most early-phase MTAP-directed trials require advanced, unresectable, metastatic, recurrent, or standard-option-exhausted disease.
Check trial inclusion/exclusion criteria	Eligibility depends on tumor type, prior therapy, ECOG performance status, organ function, measurable disease, and biopsy feasibility.
Confirm MTAP loss with acceptable assay	Many trials require homozygous MTAP deletion and/or MTAP protein loss by IHC, not just a noisy expression signal.

Final judgment

Source-specific conclusion

- Strong candidate for precision-oncology trial-screening review; not a standard-care option.
- The report is strongest as a precision-oncology trial-screening document. It does not establish a standard-of-care treatment recommendation.

11. Cross-Source Synthesis

Across all URL sections, the conclusion is consistent: the strongest actionable axis is MTAP loss with CDKN2A/CDKN2B loss, pointing to PRMT5/MAT2A-pathway trial screening. CHEK2/DDR remains a secondary but important pathway discussion. None of these sources converts the case into a proven standard-care treatment claim.

Theme	Cross-source conclusion
MTAP / PRMT5 / MAT2A	Strongest trial-screening axis. Confirm MTAP protein loss and/or homozygous deletion, then screen for MTAP-deleted solid tumor trials.
CDKN2A / CDKN2B	Supports 9p21-region biology and strengthens confidence that MTAP loss is part of a broader co-deletion pattern. Useful for validation and pathway reasoning.
CHEK2 / DDR	Supports DNA-repair vulnerability discussion, including PARP/ATR/WEE1/DNA-PK trial logic, but CHEK2 alone is not equivalent to BRCA/HRD.
TAF1 low coverage	Technical limitation only. The report could not reliably evaluate TAF1; absence of a TAF1 mutation should not be interpreted as normal biology.
Evidence maturity	Most sources are early clinical, preclinical, or trial records. The right action is confirmatory testing and trial matching, not direct treatment assertion.

Practical bottom line

- This was a high-priority molecular lead for MTAP-directed trial screening if the clinical condition and trial criteria aligned.
- The next best clinical step would have been confirmatory MTAP/CDKN2A testing and focused MTAP-deleted solid tumor trial search.
- The report should communicate the opportunity clearly while preserving the boundary: investigational rationale, not proven treatment.

12. URLs Assessed in This Appendix

Source	URL
Panome Bio — Metabolomic Logic Behind MAT2A/PRMT5 Combinations	https://panomebio.com/blog/metabolomic-logic-behind-mat2a-prmt5-combinations/
OncLive — AMG 193 Shows Preliminary Clinical Activity in MTAP-Deleted Solid Tumors	https://www.onclive.com/view/amg-193-shows-preliminary-clinical-activity-in-mtap-deleted-solid-tumors
NIH/PMC — AMG 193, a Clinical Stage MTA-Cooperative PRMT5 Inhibitor	https://pmc.ncbi.nlm.nih.gov/articles/PMC11726016/
ScienceDirect — TNG908 is a Brain-Penetrant, MTA-Cooperative PRMT5 Inhibitor	https://www.sciencedirect.com/science/article/pii/S1936523324003905
ACS — Discovery of TNG908	https://pubs.acs.org/doi/10.1021/acs.jmedchem.4c00133
Nature Communications — MAT2A Inhibitor AG-270/S095033 in Advanced Malignancies	https://www.nature.com/articles/s41467-024-55316-5
Nature / Cell Death Discovery — Combined Inhibition by PRMT5 and MAT2A Demonstrates Synthetic Lethality	https://www.nature.com/articles/s41420-025-02545-2
ClinicalTrials.gov — NCT03435250: AG-270/S095033	https://clinicaltrials.gov/study/NCT03435250
ClinicalTrials.gov — NCT05245500: MRTX1719 in Solid Tumors With MTAP Deletion	https://clinicaltrials.gov/study/NCT05245500
ClinicalTrials.gov — NCT04794699: IDE397	https://clinicaltrials.gov/study/NCT04794699
ClinicalTrials.gov — NCT07277413: IDE892 Monotherapy and Combination in MTAP-Deleted Advanced Solid Tumors	https://clinicaltrials.gov/study/NCT07277413